Astronomy - Graduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The belts and zones of Jupiter are
   (a) names for the layers of gaseous and metallic hydrogen deep within the planet.
   (b) alternating bands of rising and falling air at different latitudes.
   (c) alternating regions of charged particles in Jupiter's magnetic field.
   (d) regions of the plasma torus created by ions from Io's volcanoes

2. When traveling north from the United States into Canada you'll see the North Star (Polaris) getting ______________.
   (a) Brighter
   (b) Dimmer
   (c) Higher in the sky
   (d) Lower in the sky

3. Suppose the angular separation of two stars is smaller than the angular resolution of your eyes. How will the stars appear to your eyes?
   (a) You will not be able to see these two stars at all.
   (b) You will see two distinct stars.
   (c) The two stars will look like a single point of light.
   (d) The two stars will appear to be touching looking rather like a small dumbbell.

4. Where are the Trojan asteroids located?
   (a) in the center of the asteroid belt
   (b) on orbits that cross Earth's orbit
   (c) surrounding Jupiter
   (d) along Jupiter's orbit $60^\circ aheadofandbehindJupiter$

5. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:
   (a) asteroids would orbit with a period half that of Jupiter
   (b) asteroids would orbit with a period twice that of Jupiter
   (c) asteroids would orbit with a period twice that of Mars
   (d) A and B

True / False (10 marks)

Answer True or False

6. True or False: Mercury's atmosphere is primarily lost through volcanic heating, which continually replenishes it.
7. True or False: Organisms living deep in the oceans around seafloor volcanic vents and in hot springs most resemble the common ancestor of all life.
8. True or False: Moons significantly contribute to the stability of particles within rings, create gaps, and exert gravitational effects at ring edges.
9. True or False: Jupiter's magnetic field is approximately 20,000 times stronger than Earth's magnetic field.
10. True or False: A solar day on Planet X is approximately 100 Earth days due to its prograde rotation and orbital synchronization.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the acceleration of a truck on Mars compared to Earth, assuming no friction. What conclusions can be drawn about the role of gravitational force in this scenario?
12. Discuss the concept of black bodies in astronomy, including their definition, significance in calculating stellar radiation, and implications for understanding stellar properties.

End of Paper